

Unitary equivalence of irreducible representations

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Scope

Selected GNS cyclic subspaces assemble into a surjective isometry and then a unitary equivalence of the target models. The nonunital case first proves unit preservation; shell sums are compared on their own Hilbert spaces without assuming strong continuity of an arbitrary representation.

9 direct Cards + 20 reused prerequisites = 29 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: Unitary equivalence without an initial unitality assumption
- Key result: The cyclic sum fills every irreducible target representation

[Read this route with prerequisites](#)

Detailed route conditions

For a fixed supplied family and its once-chosen links, put $T = C^*(\pi(A), \{L_i\})$. The maps have distinct domains:

$$\iota : A \rightarrow T, \quad \rho_0 : T \rightarrow B(\mathcal{H}), \quad \rho_0 \circ \iota = \pi.$$

The selected vector states give a cyclic-sum isometry $W : \mathcal{H} \rightarrow K$. In the generated-algebra setting, root normalization and irreducibility show that the chosen isometry is surjective. The full equivalence proof uses that same underlying map after bundling it as a unitary, and additionally uses the prescribed cyclic-vector action of the links. Shell sums are compared separately on the two Hilbert spaces; strong continuity of an arbitrary representation is never assumed.

The nonunital version first proves unit preservation for the same nonzero irreducible representation. Simplicity and compact-operator exclusion are subsequent consequences. The existential unitary in a separately applied theorem need not equal a witness selected in another application.

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (4 Cards)

Level 0

The selected GNS representation of a pure-state class

A choice of representative pure states, fixed literally at a root, gives one concrete GNS representation per equivalence class.

`MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations

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Level 0

The CAR algebra as a completion of finite matrix stages

Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation.

`MathlibAnnex.CStarAlgebra.CAR.Limit`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion · The completed CAR algebra is infinite-dimensional

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Level 0

A fixed representative family of CAR shell data

One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component.

`MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation

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Level 0

The closed operator algebra generated by a representation and extra operators

Places a represented algebra and a chosen family of bounded operators in one concrete closed algebra.

`MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: An irreducible operator algebra constructed from projection shells · Reconstructing a represented unitary from its shells and residual corner

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Level 1 (5 Cards)

Level 1 The product-vector state on the completed CAR algebra Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion. MathlibAnnex.CStarAlgebra.CAR.rootState
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Purity of the completed root state
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Level 1 The completed CAR algebra is infinite-dimensional Rules out finite dimension by retaining matrix subalgebras of unbounded dimension. MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: The shell-generated algebra is not an algebra of all compact operators
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Level 1 An irreducible operator algebra constructed from projection shells Shell partial isometries with rank-one limiting defects can be completed to unitary links that join inequivalent irreducible fibers. MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_atomicShellModel
Immediate Card prerequisites: The closed operator algebra generated by a representation and extra operators Used by in this scope: The atomic-shell construction on selected pure-GNS fibers
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Level 1 Distinct GNS classes have no unitary intertwiner The class index makes the selected pure-GNS family pairwise unitarily inequivalent. MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation Immediate Card prerequisites: The selected GNS representation of a pure-state class Used by in this scope: An inequivalent GNS fiber has no residual common range · The atomic-shell construction on selected pure-GNS fibers · Assembling selected GNS cyclic subspaces into an isometric source representation Open Declaration Card Read exact source
Level 1 The decreasing root projections in the CAR completion The distinguished stage corners give a nested sequence of projections supporting the root state. MathlibAnnex.CStarAlgebra.CAR.rootFlag Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source
Level 2 (3 Cards) Level 2 Root compression becomes scalar in norm Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra. MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion Used by in this scope: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range · A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation Open Declaration Card Read exact source Level 2 Purity of the completed root state Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra. MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState Immediate Card prerequisites: The product-vector state on the completed CAR algebra Used by in this scope: The matching GNS fiber retains exactly its cyclic line · The atomic direct sum of the selected CAR representations · An inequivalent GNS fiber has no residual common range Open Declaration Card Read exact source

Level 2 The atomic-shell construction on selected pure-GNS fibers Pure-state GNS data supplies the irreducible and inequivalent fibers required by the generic shell construction. <code>MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_pureAtomicShellModel</code> Immediate Card prerequisites: Distinct GNS classes have no unitary intertwiner · An irreducible operator algebra constructed from projection shells Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra Open Declaration Card Read exact source
Level 3 (3 Cards)
Level 3 The atomic direct sum of the selected CAR representations All selected pure-GNS fibers of the completed CAR algebra act together on one Hilbert direct sum. <code>MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation</code> Immediate Card prerequisites: The selected GNS representation of a pure-state class · Purity of the completed root state Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation Open Declaration Card Read exact source
Level 3 The matching GNS fiber retains exactly its cyclic line Identifies the residual projection of a transported CAR flag in its matching pure-state representation. <code>MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_rankOne</code> Immediate Card prerequisites: Root compression becomes scalar in norm · The selected GNS representation of a pure-state class · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line Open Declaration Card Read exact source
Level 3 An inequivalent GNS fiber has no residual common range Uses compression and cyclic transport to exclude fixed vectors in every other chosen pure-state class. <code>MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero</code> Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line Open Declaration Card Read exact source

Level 4 (3 Cards)

Level 4

Assembling selected GNS cyclic subspaces into an isometric source representation

Joins mutually orthogonal pure-state cyclic copies and controls the limiting flag projections on their sum.

`MathlibAnnex.CStarAlgebra.CAR.exists_selectedAtomicCyclicIsometry`

Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations · A fixed representative family of CAR shell data

Used by in this scope: The cyclic sum fills every irreducible target representation

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Level 4

The atomic common range is one embedded GNS line

Assembles the matching and inequivalent fiber calculations in an arbitrary Hilbert direct sum.

`MathlibAnnex.CStarAlgebra.CAR.iInf_range_atomic_transportFlag_eq_span`

Immediate Card prerequisites: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range

Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion · Realizing a shell family by a faithful irreducible operator algebra

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Level 4

Reconstructing a represented unitary from its shells and residual corner

Separates a represented generator into a strong shell sum and the exact operator between its limiting fixed spaces.

`MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction`

Immediate Card prerequisites: The atomic direct sum of the selected CAR representations · The decreasing root projections in the CAR completion · A fixed representative family of CAR shell data · The closed operator algebra generated by a representation and extra operators

Used by in this scope: An irreducible target representation has a surviving fixed space

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Level 5 (2 Cards)

Level 5

Realizing a shell family by a faithful irreducible operator algebra

Constructs unitary links between pure-state summands and an irreducible algebra containing a faithful copy of CAR.

`MathlibAnnex.CStarAlgebra.CAR.exists_completedAtomicShellModel`

Immediate Card prerequisites: The atomic common range is one embedded GNS line · The atomic direct sum of the selected CAR representations · The atomic-shell construction on selected pure-GNS fibers

Used by in this scope: The ambient inclusion of the fixed shell-family target · The CAR source map into the fixed shell-family target

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Level 5

An irreducible target representation has a surviving fixed space

Rules out simultaneous disappearance of all limiting CAR flags by passing reduction through the reconstructed generators.

`MathlibAnnex.CStarAlgebra.CAR.exists_nonzero_targetFixedSpace`

Immediate Card prerequisites: Reconstructing a represented unitary from its shells and residual corner

Used by in this scope: A common root vector realizes every selected state through the generators

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Level 6 (3 Cards)

Level 6

The CAR source map into the fixed shell-family target

Restricting the codomain of the selected atomic representation gives the source map into one fixed generated target.

`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`

Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra

Used by in this scope: The shell-generated algebra is not an algebra of all compact operators · Closed-ideal simplicity of the shell-generated algebra

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Level 6

The ambient inclusion of the fixed shell-family target

The same concrete target has a literal representation on the selected atomic Hilbert space.

`MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion`

Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra

Used by in this scope: Closed-ideal simplicity of the shell-generated algebra · Unitary equivalence without an initial unitality assumption

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Level 6

A common root vector realizes every selected state through the generators

Builds a compatible family of unit vectors from a surviving residual space and identifies their CAR vector states.

MathlibAnnex.CStarAlgebra.CAR.exists_targetDefectVectorFamily

Immediate Card prerequisites: Root compression becomes scalar in norm · An irreducible target representation has a surviving fixed space

Used by in this scope: The cyclic sum fills every irreducible target representation

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Level 7 (2 Cards)

Level 7

The shell-generated algebra is not an algebra of all compact operators

Excludes an injective representation of the infinite-dimensional unital algebra onto all compact operators.

MathlibAnnex.CStarAlgebra.CAR.not_isCompactOperatorModel_shellFamilyTarget

Immediate Card prerequisites: The CAR source map into the fixed shell-family target · The completed CAR algebra is infinite-dimensional

Used by in this scope: Structural properties of one shell-generated C*-algebra

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Level 7

The cyclic sum fills every irreducible target representation

Uses shell reconstruction and residual lines to turn an isometric CAR intertwiner into a unitary equivalence on the source.

MathlibAnnex.CStarAlgebra.CAR.exists_surjective_selectedAtomicCyclicIsometry

Immediate Card prerequisites: A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation

Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion

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Level 8 (1 Card)

Level 8

Every irreducible representation is unitarily equivalent to the inclusion

Extends the unitary equivalence from the CAR source to all additional generators and the norm-closed target.

MathlibAnnex.CStarAlgebra.CAR.ambientInclusion_unitaryEquivalent

Immediate Card prerequisites: The atomic common range is one embedded GNS line · The cyclic sum fills every irreducible target representation

Used by in this scope: Closed-ideal simplicity of the shell-generated algebra · Unitary equivalence without an initial unitality assumption

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Level 9 (2 Cards)

Level 9
Unitary equivalence without an initial unitality assumption
Shows why allowing nonunital input representations does not enlarge the irreducible representation class of the shell-generated algebra.
<code>MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion_unitaryEquivalent_nonUnital</code>
Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · Every irreducible representation is unitarily equivalent to the inclusion
Used by in this scope: Structural properties of one shell-generated C*-algebra
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Level 9
Closed-ideal simplicity of the shell-generated algebra
Uses a faithful irreducible representation in the unique unitary-equivalence class to rule out proper nonzero closed two-sided ideals.
<code>MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_shellFamilyTarget</code>
Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · Every irreducible representation is unitarily equivalent to the inclusion · The CAR source map into the fixed shell-family target
Used by in this scope: Structural properties of one shell-generated C*-algebra
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Level 10 (1 Card)

Level 10
Structural properties of one shell-generated C*-algebra
Collects the proved properties of a single generated algebra, keeping the shell-family input separate from its consequences.
<code>MathlibAnnex.CStarAlgebra.CAR.shellFamilyEndpoint</code>
Immediate Card prerequisites: The shell-generated algebra is not an algebra of all compact operators · Closed-ideal simplicity of the shell-generated algebra · Unitary equivalence without an initial unitality assumption
Used by in this scope: None in this selected scope
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